

Assingment —2

Task 1: Statistical Summary for Survey Data

Prompt: generate a Python function that reads a list of numbers and calculates the mean, minimum, and maximum values. **Code:**

```
def calculate_statistics(numbers):  
    if not numbers:  
        return None, None, None # Return None for mean, min, and max if the list is empty  
  
    mean_value = sum(numbers) / len(numbers)  
    min_value = min(numbers)    max_value =  
    max(numbers)
```

```
    return mean_value, min_value, max_value #exapmple
```

usage:

```
numbers = [10, 20, 30, 40, 50] mean, minimum, maximum =  
calculate_statistics(numbers) print(f"Mean: {mean}, Minimum:  
{minimum}, Maximum: {maximum}")
```

output:



```
PS C:\Users\kandi\OneDrive\Desktop\assignment-2> & C:\Users\kandi\AppData\Local\python\pythoncore-3.14-64\python.exe c:/Users/kandi/OneDrive/Desktop/assignment-2/task-1.py  
Mean: 30.0, Minimum: 10, Maximum: 50
```

Task 2: Armstrong Number – AI Comparison

Prompt: Generate an Armstrong number checker using Given a number, check if it is an Armstrong number or not. Compare their outputs, logic style, and clarity.

Code:

```
def is_armstrong_number(n):  
    # Convert the number to a string to easily iterate over each digit  
    digits = str(n)    num_digits = len(digits)  
  
    # Calculate the sum of each digit raised to the power of the number of digits  
    armstrong_sum = sum(int(digit) ** num_digits for digit in digits)  
  
    # Check if the calculated sum is equal to the original number  
    return armstrong_sum == n # Example usage: number = 153    if  
is_armstrong_number(number):  
    print(f"{number} is an Armstrong number.") else:  
    print(f"{number} is not an Armstrong number.") output:
```

```
PS C:\Users\kandi\OneDrive\Desktop\assignment-2> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/kandi/OneDrive/Desktop/assignment-2/task-2.py  
153 is an Armstrong number.
```

Task 3: Leap Year Validation Using Cursor AI

Prompt: generate a Python program that checks whether a given year is a leap year.also give two different methods to check for leap year and compare their outputs, logic style, and clarity.

Code:

```
# Method 1: Using the standard leap year rules def  
is_leap_year_method1(year):    if (year % 4 == 0 and year %  
100 != 0) or (year % 400 == 0):  
    return True  
else:  
    return False  
  
# Example usage:
```

```

year = 2020 if
is_leap_year_method1(year):
print(f"{year} is a leap year.") else:
print(f"{year} is not a leap year.")

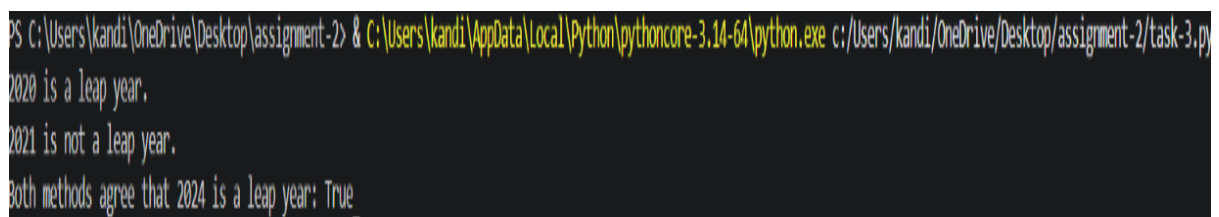
# Method 2: Using a more concise approach def
is_leap_year_method2(year):
    return year % 4 == 0 and (year % 100 != 0 or year % 400 == 0)
# Example usage: year = 2021    if is_leap_year_method2(year):
    print(f"{year} is a leap year.")
else:    print(f"{year} is not a leap
year.") #example usage:

year = 2024 if is_leap_year_method1(year) ==
is_leap_year_method2(year):
    print(f"Both methods agree that {year} is a leap year: {is_leap_year_method1(year)}")

```

output

:



```

PS C:\Users\kandi\OneDrive\Desktop\assignment-2> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/kandi/OneDrive/Desktop/assignment-2/task-3.py
2020 is a leap year.
2021 is not a leap year.
Both methods agree that 2024 is a leap year: True

```

Task 4: Student Logic (Odd/Even Sum)

Prompt: Write a Python program that calculates the sum of odd and even numbers in a tuple.

Code:

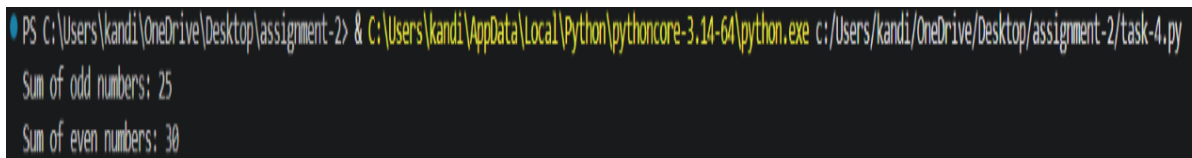
```
def sum_odd_even(numbers):  
    odd_sum = sum(x for x in numbers if x % 2 != 0)  
    even_sum = sum(x for x in numbers if x % 2 == 0)    return  
    odd_sum, even_sum
```

Example usage:

```
tuple_numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10) odd_total,  
even_total = sum_odd_even(tuple_numbers) print(f"Sum  
of odd numbers: {odd_total}") print(f"Sum of even  
numbers: {even_total}")
```

output

:



```
PS C:\Users\kandi\OneDrive\Desktop\assignment-2> & C:\Users\kandi\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/kandi/OneDrive/Desktop/assignment-2/task-4.py  
Sum of odd numbers: 25  
Sum of even numbers: 30
```